RATNAJIT DHAR

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ratnajit-dhar | ratnajit-dhar | Portfolio - Ratnajit Dhar | Ratnajit Dhar

Education

- **Chittagong University of Engineering & Technology** *March 2022 - July 2026*
  BSc. in Computer Science & Engineering
  - GPA: 3.86/4.00

Experience

- **Tech Solutions Architect** *June 2024 - January 2026*
  Green Lead Remote
  - Worked on a project named Re-Energized Bangladesh on the Climate Intelligence Technology Team which involves creating a renewable energy dashboard in Excel for Bangladesh.
  - Developed a climate based chatbot using RAG.

Skills

- **Programming Languages:** Python, C, C++
- **Web Technologies:** HTML, CSS, Javascript
- **Databases & Vector Stores:** MySQL, PostgreSQL, FAISS, Pinecone
- **Frameworks & Libraries:** PyTorch, TensorFlow, FastAPI, Flask, Streamlit, LangChain, LangGraph, LangSmith, Hugging Face
- **Tools:** GitHub, LaTeX

Undergraduate Thesis

Title: Chain-of-Thought Enhanced Multimodal Question Answering System for Computer Science Domains Using Vision-Language Models
Supervisor: Dr. Mahfuzulhoq Chowdhury, Professor, CSE, CUET
Description: Developed a Visual Question Answering (VQA) system to answer visual questions in computer science domains. The system uses Chain-of-Thought reasoning for explainable responses. Evaluation was conducted using Accuracy, Binary LAVE and LLM-As-A-Judge metrics.

Projects

- **SupportRAG - Enterprise IT Support RAG**
  Tools: LangGraph, FastAPI, Pinecone, OpenAI API, Tavily
  - Developed an Agentic RAG system using LangGraph that routes queries to an internal knowledge base or web search based on the question and retrieved evidence.
  - Built a FastAPI backend with document ingestion, semantic retrieval using Pinecone, and source-grounded response generation.
- **ResGent - Multi Agent Research System**
  Tools: LangChain, Tavily, BeautifulSoup, OpenRouter
  - Developed a multi-agent research system that automates web search, content extraction, and structured report generation using LangChain and LLMs.
  - Integrated search and URL scraping tools with four specialized agents: Search, Reader, Writer, and Critic.
- **Spam Classification with MLOps**
  Tools: Python,MLFlow, DVC
  - Developed a spam classification system using machine learning with an end-to-end MLOps pipeline, including data ingestion, preprocessing, feature extraction, model training, and evaluation.
  - Ensured reproducibility and scalability by integrating DVC for data/pipeline versioning and MLflow for experiment tracking and model management.
- **CGPA Calculator For CUET Students**
  Tools: Flask, Pandas, HTML, CSS, Google Colab
  - Developed a web application to automate CGPA calculation for CUET students with a user-friendly interface for quick grade input.

◦ Built the backend using Flask to ensure efficient data processing and instant result generation.

- **CUET Thesis & Project Portal**

Tools: Firebase

◦ Developed the backend for a centralized repository of student theses and projects using Firebase for data storage and user authentication.

◦ Designed the system architecture to enable efficient access, management, and sharing of academic work.

Awards & Achievements

Finalist - Team 823_OLT — BUET DL Sprint 4.0 *2026*
Champion - Team CUET_RSCube — DU AI Challenge - National Datathon *2025*
Bronze Award Winner - Team CUET_R2S4 — Bangladesh Blockchain Olympiad - AI Category *2025*
1st Runner-up — Intra CUET ML Contest - IEEE CS CUET Student Branch Chapter *2025*
4th Place — Shared Task on Misogyny Meme Detection - DravidianLangTech@NAACL *2025*
Honorable Mention — ICPC Asia Dhaka Regional Contest *2024*
Specialist — Codeforces *2024*

Publications

C=Conference, J=Journal, S=In Submission, T=Thesis, P=Preprint

[S.2] **Ratnajit Dhar**, Arpita Mallik. **Gap-Aware Windowing and Task-Specific Preprocessing for Bengali Long-Form ASR and Diarization**. In *29th International Conference on Computer and Information Technology (ICCIT).*

[S.1] A.F.M. Farhad, Sabik Aftahee, Arpita Mallik, **Ratnajit Dhar**, Jawadul Karim, Nahiyan Bin Noor, Ishmam Ahmed SolaIman. **Assessing the Reliability of Large Language Models in the Bengali Legal Context: A Comparative Evaluation Using LLM-as-Judge and Legal Experts**. In *29th International Conference on Computer and Information Technology (ICCIT).*

[C.3] Sajib Bhattacharjee, **Ratnajit Dhar**, Kawsar Ahmed, Mohammed Moshiul Hoque. **Binary_Bunch at AraHealthQA Track 1: Arabic Mental Health Q&A Classification Using Data Augmentation and Transformer Models**. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks, pages 164–175, Suzhou, China. Association for Computational Linguistics.*

[C.2] **Ratnajit Dhar**, Arpita Mallik. **CUET-823 at MAHED 2025 Shared Task: Large Language Model-Based Framework for Emotion, Offensive, and Hate Detection in Arabic**. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks, pages 620–625, Suzhou, China. Association for Computational Linguistics.*

[C.1] Arpita Mallik, **Ratnajit Dhar**, Udoy Das, Momtazul Arefin Labib, Samia Rahman, Hasan Murad. **CUET-823@DravidianLangTech 2025: Shared Task on Multimodal Misogyny Meme Detection in Tamil Language**. In *Proceedings of the Fifth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages*.

Competitive Programming Experience

- **Specialist @ Codeforces -** Max Rating: 1525
- **Honorable Mention -** ICPC Asia Dhaka Regional Contest 2024

Leadership Experience

Chair *July 2025 - August 2026*
IEEE Computer Society CUET Student Branch Chapter

Joint Organizing Secretary *July 2025 - August 2026*
CUET Computer Club

Joint General Secretary *July 2025 - August 2026*
Joyoddhoney, CUET

Head of Finance *January 2025 - February 2025*
Hult Prize @ CUET